

Labsheet - 3

Statistical Analysis and Hypothesis Testing using Python

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In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import ttest_ind
from scipy.stats import f_oneway
import statsmodels.api as sm
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

# Step 1: Load dataset
file_path = "WA_Fn-UseC_-HR-Employee-Attrition-selected-columns.csv"
df = pd.read_csv(file_path)
print("Dataset loaded successfully!")
print("\nDataset Shape:")
print(df.shape)
print("\nFirst 5 Records:")
print(df.head())

# Step 2: Basic information
print("\nDataset Information:")
df.info()
print("\nMissing Values:")
print(df.isnull().sum())
print("\nColumn Names:")
print(df.columns.tolist())

# Step 3: Descriptive Statistics
numerical_columns = [
    "Age",
    "DailyRate",
    "DistanceFromHome",
    "Education",
    "EmployeeCount",
    "EmployeeNumber"
]
print("\nMean:")
print(df[numerical_columns].mean())
print("\nMedian:")
print(df[numerical_columns].median())
print("\nMode:")
print(df[numerical_columns].mode().iloc[0])
print("\nVariance:")
print(df[numerical_columns].var())
print("\nStandard Deviation:")
print(df[numerical_columns].std())
print("\nComplete Descriptive Statistics:")
print(df[numerical_columns].describe())

# Step 4: Pearson Correlation
correlation = df[numerical_columns].corr()
print("\nPearson Correlation Matrix:")
print(correlation)
plt.figure(figsize=(10, 7))
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sns.heatmap(
    correlation,
    annot=True,
    cmap="coolwarm",
    fmt=".2f"
)
plt.title("Pearson Correlation Matrix")
plt.tight_layout()
plt.show()

# Step 5: Hypothesis
print("\nHypothesis Testing")
print("H0: There is no significant difference in Age")
print("    between employees who leave and employees who stay.")
print("\nH1: There is a significant difference in Age")
print("    between employees who leave and employees who stay.")

# Step 6: Independent Sample T-Test
age_left = df[
    df["Attrition"] == "Yes"
]["Age"]
age_stayed = df[
    df["Attrition"] == "No"
]["Age"]
t_statistic, p_value = ttest_ind(
    age_left,
    age_stayed,
    equal_var=False
)
print("\nIndependent Sample T-Test")
print("Mean Age - Employees who left:")
print(age_left.mean())
print("Mean Age - Employees who stayed:")
print(age_stayed.mean())
print("T-Statistic:", t_statistic)
print("P-Value:", p_value)
alpha = 0.05
if p_value < alpha:
    print("\nResult: Reject H0.")
    print("There is a statistically significant difference")
    print("in Age between employees who left and stayed.")
else:
    print("\nResult: Fail to reject H0.")
    print("There is no statistically significant difference")
    print("in Age between employees who left and stayed.")

# Step 7: One-Way ANOVA
print("\nOne-Way ANOVA")
print("Testing Age differences across Departments.")
department_groups = []
departments = df["Department"].unique()
for department in departments:
    group = df[
        df["Department"] == department
    ]["Age"]
    department_groups.append(group)
f_statistic, anova_p_value = f_oneway(
    *department_groups
)
print("Departments:")

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print(departments)
print("F-Statistic:", f_statistic)
print("P-Value:", anova_p_value)
if anova_p_value < alpha:
    print("\nResult: Reject H0.")
    print("There is a significant difference in Age")
    print("among the different departments.")
else:
    print("\nResult: Fail to reject H0.")
    print("There is no significant difference in Age")
    print("among the different departments.")

# Department vs Age visualization
plt.figure(figsize=(8, 6))
sns.boxplot(
    data=df,
    x="Department",
    y="Age"
)
plt.title("Age Distribution Across Departments")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()

# Step 8: Simple Linear Regression
# Independent variable = DistanceFromHome
# Dependent variable = DailyRate
X = df[["DistanceFromHome"]]
y = df["DailyRate"]
model = LinearRegression()
model.fit(X, y)
y_pred = model.predict(X)
slope = model.coef_[0]
intercept = model.intercept_
r2 = r2_score(
    y,
    y_pred
)
print("\nLinear Regression")
print("Independent Variable: DistanceFromHome")
print("Dependent Variable: DailyRate")
print("Slope:", slope)
print("Intercept:", intercept)
print("R2 Score:", r2)
# Regression Plot
plt.figure(figsize=(8, 6))
plt.scatter(
    X,
    y,
    alpha=0.5
)
plt.plot(
    X,
    y_pred
)
plt.xlabel("Distance From Home")
plt.ylabel("Daily Rate")
plt.title(
    "Linear Regression: Distance From Home vs Daily Rate"
)

```

```
plt.tight_layout()
plt.show()

# Step 9: Statsmodels Regression
X_stats = sm.add_constant(X)
regression_model = sm.OLS(
    y,
    X_stats
).fit()
print("\nStatsmodels Regression Summary:")
print(regression_model.summary())
print("\nRegression Coefficients:")
print(regression_model.params)
print("\nP-Values:")
print(regression_model.pvalues)
print("\n95% Confidence Intervals:")
print(regression_model.conf_int())
print("\nR2 Score:")
print(regression_model.rsquared)

# Step 10: Final Summary
print("\nFinal Summary")
print("1. Descriptive statistics were calculated.")
print("2. Pearson correlation analysis was performed.")
print("3. Independent Sample T-Test was performed.")
print("4. One-Way ANOVA was performed.")
print("5. Linear Regression was performed.")
print("6. P-values were analyzed.")
print("7. Confidence intervals were calculated.")
print("8. R2 score was calculated.")
print("\nStatistical analysis helps identify")
print("relationships, differences, and trends")
print("in real-world datasets.")
print("\nExperiment No. 3 completed successfully!")
```

Dataset loaded successfully!

Dataset Shape:
(1470, 10)

First 5 Records:

	Age	Attrition	BusinessTravel	DailyRate	Department	\
0	41	Yes	Travel_Rarely	1102		Sales
1	49	No	Travel_Frequently	279	Research & Development	
2	37	Yes	Travel_Rarely	1373	Research & Development	
3	33	No	Travel_Frequently	1392	Research & Development	
4	27	No	Travel_Rarely	591	Research & Development	

	DistanceFromHome	Education	EducationField	EmployeeCount	EmployeeNumber
0	1	2	Life Sciences	1	1
1	8	1	Life Sciences	1	2
2	2	2	Other	1	4
3	3	4	Life Sciences	1	5
4	2	1	Medical	1	7

Dataset Information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 1470 entries, 0 to 1469

Data columns (total 10 columns):

#	Column	Non-Null Count	Dtype
0	Age	1470 non-null	int64
1	Attrition	1470 non-null	object
2	BusinessTravel	1470 non-null	object
3	DailyRate	1470 non-null	int64
4	Department	1470 non-null	object
5	DistanceFromHome	1470 non-null	int64
6	Education	1470 non-null	int64
7	EducationField	1470 non-null	object
8	EmployeeCount	1470 non-null	int64
9	EmployeeNumber	1470 non-null	int64

dtypes: int64(6), object(4)

memory usage: 115.0+ KB

Missing Values:

Age	0
Attrition	0
BusinessTravel	0
DailyRate	0
Department	0
DistanceFromHome	0
Education	0
EducationField	0
EmployeeCount	0
EmployeeNumber	0

dtype: int64

Column Names:

['Age', 'Attrition', 'BusinessTravel', 'DailyRate', 'Department', 'DistanceFromHome', 'Education', 'EducationField', 'EmployeeCount', 'EmployeeNumber']

Mean:

Age	36.923810
DailyRate	802.485714
DistanceFromHome	9.192517

Education 2.912925
 EmployeeCount 1.000000
 EmployeeNumber 1024.865306
 dtype: float64

Median:

Age 36.0
 DailyRate 802.0
 DistanceFromHome 7.0
 Education 3.0
 EmployeeCount 1.0
 EmployeeNumber 1020.5
 dtype: float64

Mode:

Age 35.0
 DailyRate 691.0
 DistanceFromHome 2.0
 Education 3.0
 EmployeeCount 1.0
 EmployeeNumber 1.0
 Name: 0, dtype: float64

Variance:

Age 83.455049
 DailyRate 162819.593737
 DistanceFromHome 65.721251
 Education 1.048914
 EmployeeCount 0.000000
 EmployeeNumber 362433.299749
 dtype: float64

Standard Deviation:

Age 9.135373
 DailyRate 403.509100
 DistanceFromHome 8.106864
 Education 1.024165
 EmployeeCount 0.000000
 EmployeeNumber 602.024335
 dtype: float64

Complete Descriptive Statistics:

	Age	DailyRate	DistanceFromHome	Education	EmployeeCount \
count	1470.000000	1470.000000	1470.000000	1470.000000	1470.0
mean	36.923810	802.485714	9.192517	2.912925	1.0
std	9.135373	403.509100	8.106864	1.024165	0.0
min	18.000000	102.000000	1.000000	1.000000	1.0
25%	30.000000	465.000000	2.000000	2.000000	1.0
50%	36.000000	802.000000	7.000000	3.000000	1.0
75%	43.000000	1157.000000	14.000000	4.000000	1.0
max	60.000000	1499.000000	29.000000	5.000000	1.0

EmployeeNumber

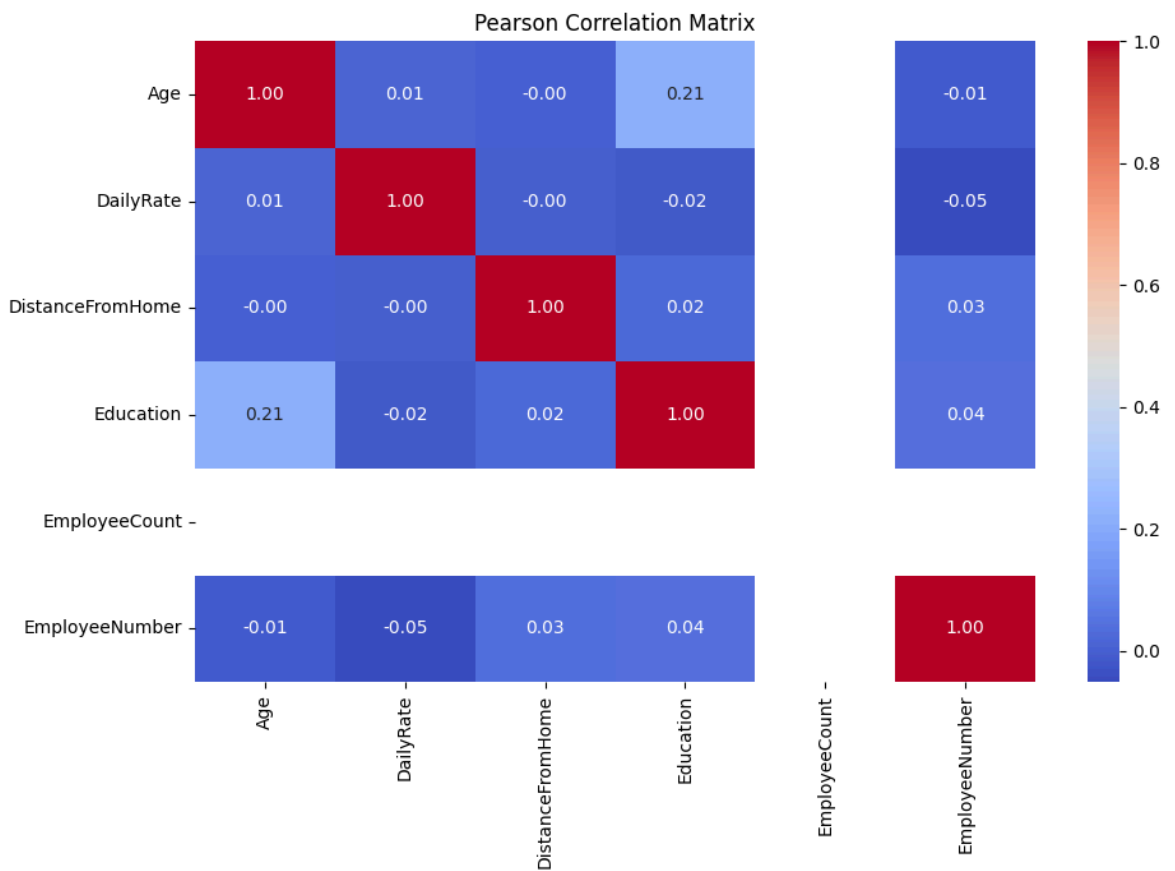
count	1470.000000
mean	1024.865306
std	602.024335
min	1.000000
25%	491.250000
50%	1020.500000
75%	1555.750000

max 2068.000000

Pearson Correlation Matrix:

	Age	DailyRate	DistanceFromHome	Education	\
Age	1.000000	0.010661	-0.001686	0.208034	
DailyRate	0.010661	1.000000	-0.004985	-0.016806	
DistanceFromHome	-0.001686	-0.004985	1.000000	0.021042	
Education	0.208034	-0.016806	0.021042	1.000000	
EmployeeCount	NaN	NaN	NaN	NaN	
EmployeeNumber	-0.010145	-0.050990	0.032916	0.042070	

	EmployeeCount	EmployeeNumber
Age	NaN	-0.010145
DailyRate	NaN	-0.050990
DistanceFromHome	NaN	0.032916
Education	NaN	0.042070
EmployeeCount	NaN	NaN
EmployeeNumber	NaN	1.000000



Hypothesis Testing

H₀: There is no significant difference in Age
between employees who leave and employees who stay.

H₁: There is a significant difference in Age
between employees who leave and employees who stay.

Independent Sample T-Test

Mean Age - Employees who left:

33.607594936708864

Mean Age - Employees who stayed:

37.561232765612324

T-Statistic: -5.82801185398895

P-Value: 1.3797600649439775e-08

Result: Reject H₀.

There is a statistically significant difference
in Age between employees who left and stayed.

One-Way ANOVA

Testing Age differences across Departments.

Departments:

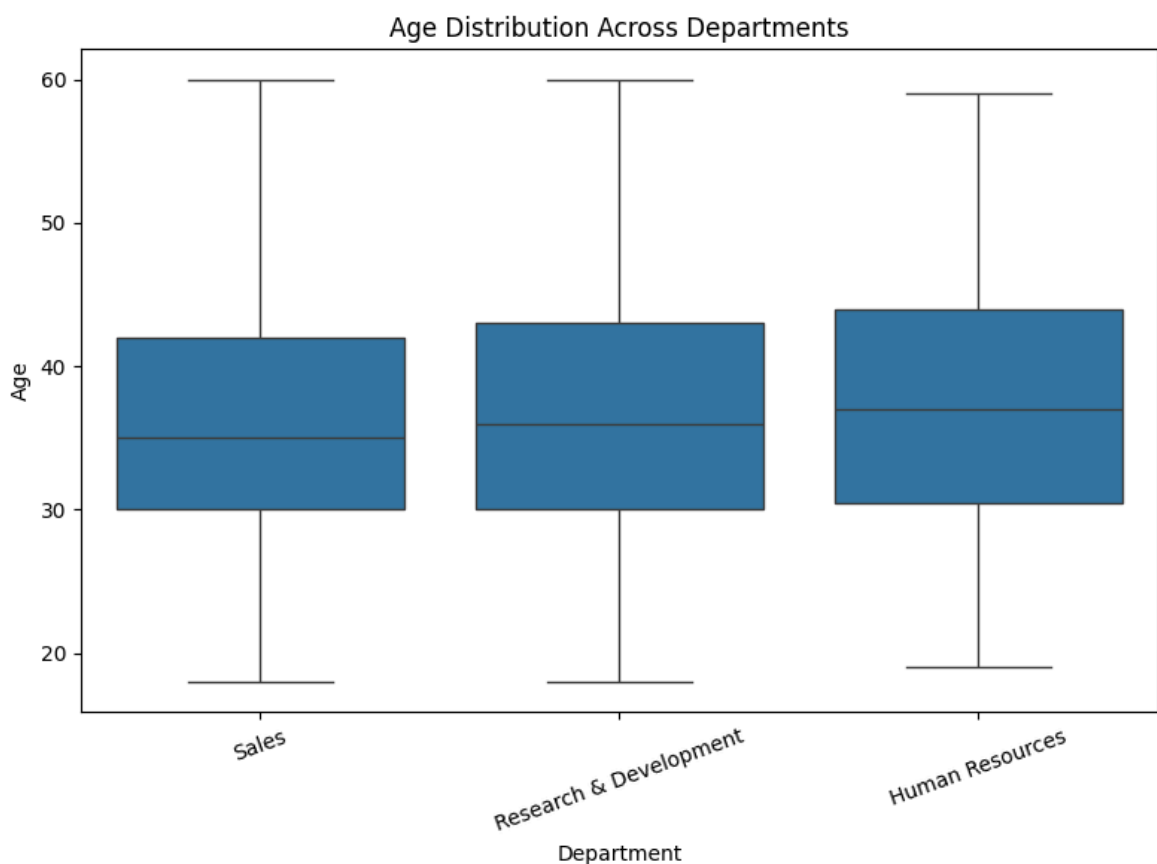
['Sales' 'Research & Development' 'Human Resources']

F-Statistic: 0.7655031924976903

P-Value: 0.46528552999349515

Result: Fail to reject H₀.

There is no significant difference in Age
among the different departments.



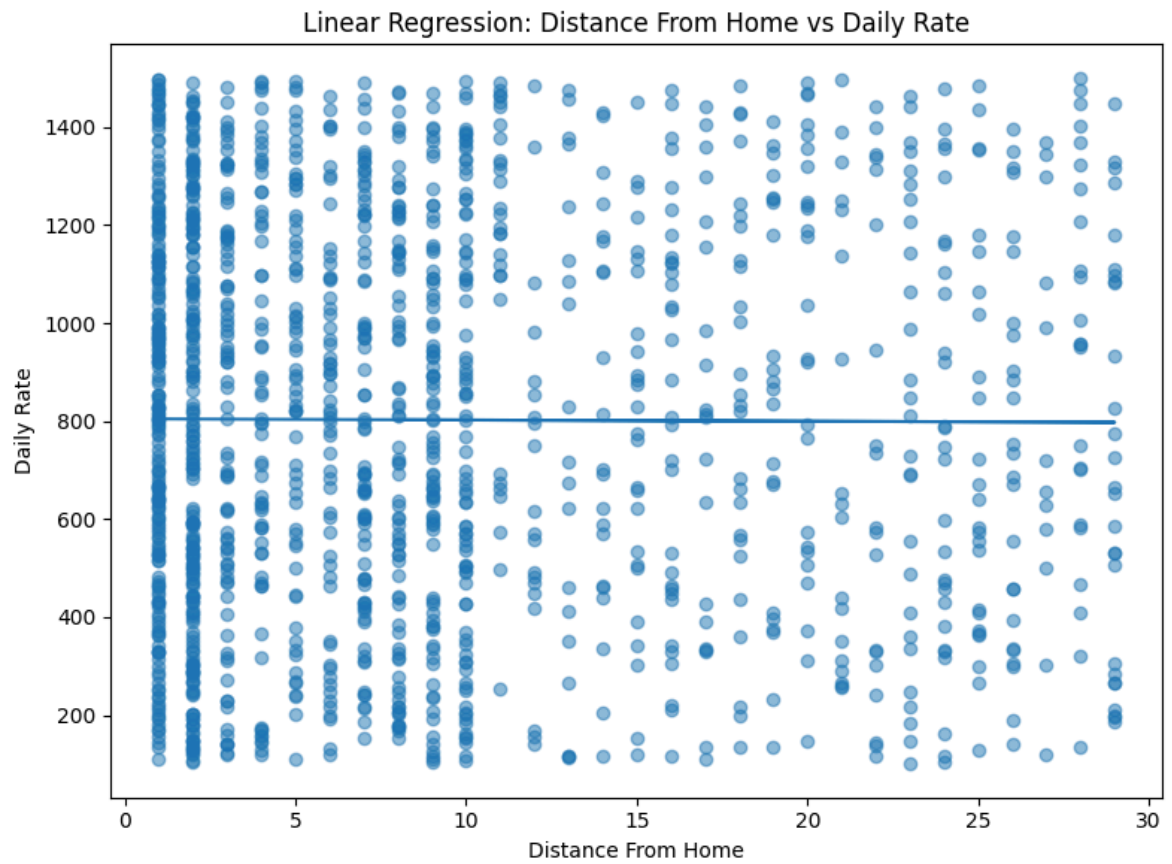
Linear Regression

Independent Variable: DistanceFromHome

Dependent Variable: DailyRate

Slope: -0.2481389696358105

Intercept: 804.766735984142

 R^2 Score: $2.485358851878061e-05$ 

Statsmodels Regression Summary:

OLS Regression Results

```

=====
Dep. Variable:          DailyRate    R-squared:                0.000
Model:                  OLS          Adj. R-squared:           -0.001
Method:                 Least Squares  F-statistic:              0.03649
Date:                   Mon, 07 Sep 2026  Prob (F-statistic):       0.849
Time:                   20:27:34      Log-Likelihood:           -10906.
No. Observations:       1470         AIC:                     2.182e+04
Df Residuals:           1468         BIC:                     2.183e+04
Df Model:                1
Covariance Type:        nonrobust
=====
===
               coef      std err          t      P>|t|      [0.025      0.9
75]
-----
---
const          804.7667      15.920      50.551      0.000      773.539      835.
995
DistanceFromHome -0.2481       1.299     -0.191      0.849      -2.796       2.
300
=====
Omnibus:            1173.556    Durbin-Watson:           2.082
Prob(Omnibus):       0.000    Jarque-Bera (JB):         88.680
Skew:                -0.003    Prob(JB):                 5.54e-20
Kurtosis:            1.797    Cond. No.                  18.6
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Regression Coefficients:

```

const          804.766736
DistanceFromHome -0.248139
dtype: float64

```

P-Values:

```

const          1.976263e-323
DistanceFromHome 8.485418e-01
dtype: float64

```

95% Confidence Intervals:

```

               0               1
const          773.538836  835.994636
DistanceFromHome -2.796367  2.300089

```

R² Score:

2.485358851878061e-05

Final Summary

1. Descriptive statistics were calculated.
2. Pearson correlation analysis was performed.
3. Independent Sample T-Test was performed.
4. One-Way ANOVA was performed.
5. Linear Regression was performed.
6. P-values were analyzed.
7. Confidence intervals were calculated.
8. R² score was calculated.

Statistical analysis helps identify relationships, differences, and trends in real-world datasets.

Experiment No. 3 completed successfully!

Question

1. What is the difference between descriptive statistics and inferential statistics?

Descriptive statistics summarize and describe data.

Inferential statistics use sample data to draw conclusions about a population.

Descriptive	Inferential
Summarizes data.	Draws conclusions.
Mean, median, mode.	t-test, ANOVA.
Describes observed data.	Tests hypotheses.

2. Explain the concepts of the Null Hypothesis (H₀) and Alternative Hypothesis (H₁).

Null Hypothesis (H₀): Assumes there is no significant difference or relationship.

Alternative Hypothesis (H₁): Assumes there is a significant difference or relationship.

Example:

H₀: There is no significant difference between two groups.

H₁: There is a significant difference between two groups.

3. What is a p-value? How is it used to make statistical decisions?

A **p-value** indicates how compatible the observed result is with the Null Hypothesis.

- **p-value < 0.05** → Reject H₀.
- **p-value ≥ 0.05** → Fail to reject H₀.

A small p-value provides stronger evidence against H₀.

4. Differentiate between a t-test and ANOVA. In which situations is each test applied?

t-test	ANOVA
Compares two groups.	Compares three or more groups.
Uses t-statistic.	Uses F-statistic.

t-test**ANOVA**

Example: two employee groups. Example: multiple departments.

5. What does the Pearson correlation coefficient indicate? What are its possible values?

The **Pearson correlation coefficient (r)** measures the strength and direction of a linear relationship between two numerical variables.

Its value ranges from **-1 to +1**.

- **+1** → Perfect positive correlation.
 - **0** → No linear correlation.
 - **-1** → Perfect negative correlation.
-

6. Explain the significance of R^2 (Coefficient of Determination) in Linear Regression.

R^2 shows the proportion of variation in the dependent variable explained by the regression model.

For example:

$R^2 = 0.80$ means the model explains about **80% of the variation** in the dependent variable for the fitted data.

A value closer to **1** generally indicates greater explanatory power.

7. Why is statistical analysis important before applying machine learning algorithms?

Statistical analysis helps to:

- Understand the dataset.
 - Find relationships between variables.
 - Detect unusual values.
 - Identify important features.
 - Check data assumptions.
 - Improve data preprocessing and model development.
-

8. What assumptions should be satisfied before performing a t-test or ANOVA?

Important assumptions include:

- Observations should be independent.
 - Data should be approximately normally distributed within groups.
 - Variances should be reasonably similar for standard tests.
 - The dependent variable should be numerical.
-

9. How can regression analysis help organizations in forecasting and decision-making?

Regression analysis helps organizations to:

- Identify relationships between variables.
- Predict future values.
- Forecast trends.
- Support planning.
- Make data-driven decisions.

For example, regression can be used to predict **sales, prices, or demand**.

10. Give two real-world applications where hypothesis testing is commonly used in data analytics.

1. Business and Marketing

Used to determine whether a new advertising campaign significantly affects sales or customer engagement.

2. Healthcare

Used to determine whether a treatment produces a significant difference compared with a comparison group.

Other applications include:

- Quality control.
- Finance.
- Education.
- A/B testing.